

PROJECT PROPOSAL

AI-Powered Enquiry & Enrollment System

Q&M Dental Group (Singapore) Limited

2-Day Basic Certificate in Dental Assisting

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Submitted to
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1. Executive Summary

Q&M Dental Group's 2-Day Basic Certificate in Dental Assisting programme currently relies entirely on manual WhatsApp-based communication for enquiries, enrollment, invoicing, and payment processing. This creates significant operational bottlenecks, increases the risk of human error, and limits the programme's scalability.

This proposal presents a single integrated project — the AI-Powered Enquiry & Enrollment System — designed to automate the end-to-end participant journey from first enquiry to receipt issuance. The system is accessed entirely through a single WhatsApp conversation: a shared Intent Router (Claude AI) detects whether each message is an enquiry, an enrollment request, or a payment submission, and automatically routes it to the appropriate processing module. The system is structured in three functional modules plus the shared Intent Router, deliverable within 12 weeks at a total budget of S\$9,000 – S\$10,000 with MSG Grant.

2. Problem Statement

Operations Manager at Q&M Dental Group, identified two distinct categories of manual work consuming significant staff time:

Group 1 — Repetitive Enquiries

A significant amount of staff time is spent manually responding to the same questions via WhatsApp, including:

- Eligibility for SkillsFuture funding, Mid-Career SkillsFuture Credits, and other subsidies
- Course benefits and learning outcomes
- Whether participants without experience or certificates can attend
- Whether job opportunities are available after course completion
- Follow-up reminders for participants who do not respond after initial contact

Group 2 — Enrollment & Payment Administration

The enrollment and payment process involves multiple sequential manual tasks, each performed individually per participant:

- Sending course information templates individually via WhatsApp
- Explaining payment modes such as SkillsFuture claims or PayNow
- Assisting participants in checking SkillsFuture credit balances using Singpass
- Collecting participant details: full name, NRIC, date of birth, email address, and preferred course date
- Generating accurate invoices manually using Excel files
- Printing, signing, and emailing participants PDF invoices and SkillsFuture claim instructions
- Sending company UEN and QR codes for PayNow transactions
- Verifying payment screenshots submitted by participants
- Updating payment records manually for accounting reconciliation
- Coordinating with accountants for receipt issuance and credit note processing for unpaid participants

These manual processes create operational inefficiencies, increase the risk of human error, delay response times, and reduce scalability when handling a large volume of enquiries and enrollments.

3. Proposed Solution — Three-Module System

The proposed system addresses all requirements through three integrated modules, unified by a shared WhatsApp Intent Router. The Intent Router is the single entry point for all customer communication — it analyses each inbound WhatsApp message and automatically routes it to the appropriate module. Each module corresponds directly to the problem areas identified:

Module A — AI Chatbot & Lead Nurturing

WhatsApp Business API + Claude AI | Addresses: Group 1 requirements

Module A deploys an AI-powered WhatsApp chatbot that handles all repetitive enquiries instantly and without staff involvement. Built on the WhatsApp Business API and powered by the LLM API as its natural language understanding engine, the bot provides accurate, consistent answers around the clock.

Features & Capabilities

- Instant AI-generated answers to all FAQ categories: SkillsFuture funding eligibility, Mid-Career Credits, course benefits, entry requirements, and post-course job opportunities
- Quick-reply menu presented on first contact for common topics, reducing free-text input from participants
- Structured lead capture conversation flow: collects name, NRIC, date of birth, email address, and preferred course date through guided WhatsApp messages
- Automated follow-up reminder sequence: messages sent at 24-hour, 72-hour, and 7-day intervals for leads who have not responded after initial contact

- Human escalation flag: complex, sensitive, or unrecognised queries are flagged and routed to a staff inbox for manual follow-up
- SkillsFuture balance guidance: the bot sends participants a direct link and step-by-step instructions to check their own balance via the official MySkillsFuture portal — no government system access required

Note: SkillsFuture balance checking via Singpass is intentionally excluded from automation. Direct integration with government identity systems raises data sovereignty and national security considerations. The chatbot instead guides participants to check independently via the official portal, which is the approach used by most accredited training providers in Singapore.

Module B — Enrollment Invoice and Payment Pipeline

Web form + PDF engine + Email delivery | Addresses: enrollment & invoice requirements

Module B replaces the current manual enrollment process with a fully automated pipeline. When a customer's inbound WhatsApp message is classified as enrollment intent by the shared Intent Router, the automation triggers the full enrollment workflow: data validation, PDF invoice generation, SkillsFuture and PayNow instructions delivery via email, and WhatsApp confirmation. When the student completes payment, they send their payment screenshot via WhatsApp, which the Intent Router classifies as payment submission intent and routes directly to Module C for automated verification.

Features & Capabilities

- Chatbot or Whatsapp capture student data already earlier in Module.
- Automatic PDF invoice generation the current Excel-based manual process; invoice includes participant details, course date, fee breakdown, and SkillsFuture subsidy amount
- Automated email delivery to participant containing: the signed PDF invoice, SkillsFuture claim instructions, PayNow QR code, and company UEN
- All enrollment records saved to a central database with real-time status tracking: Enquiry, Enrolled, Invoice Sent, Awaiting Payment, Paid, Receipt Issued
- Admin view showing all enrollments and current status, accessible to Q&M staff via a simple web dashboard

Module C — Payment Verification & Accounts

AI image verification + Accountant portal | Addresses: payment & accounting requirements

Module C closes the final gap in the workflow by automating payment screenshot verification and providing a dedicated portal for the accounting team. This module eliminates the manual screenshot review, removes ad-hoc accounting coordination, and enables one-click receipt and credit note generation.

Features & Capabilities

- Participants submit their PayNow payment screenshot via WhatsApp or email

- AI-assisted payment verification: the system reads the submitted screenshot and checks whether the transferred amount matches the invoice; matched payments are auto-confirmed, mismatches are flagged for staff review
- Automatic update of the enrollment record to Paid status upon confirmed payment — no manual data entry required
- Accountant portal: a password-protected web interface showing all enrollments filtered by payment status; the accountant can see pending receipts, overdue accounts, and credit note requests in one view. Or can periodically automatically report can be done.
- Auto receipt generation: the AI system generates a PDF receipt and emails it to the participant directly.
- Credit note workflow: unpaid or cancelled enrollments are flagged with a credit note request; the accountant reviews and approves with a single action

Note: Payment screenshot verification is AI-assisted, not fully autonomous. All mismatches and edge cases are routed to staff for final confirmation before a record is marked as paid. This ensures accuracy is maintained while eliminating the time spent on straightforward manual checks.

4. Requirements Coverage Matrix

The table below maps every requirement from the original problem statement to the module that addresses it.

Requirement	Status	Module
Group 1 — Repetitive Enquiries		
SkillsFuture funding eligibility FAQs	Fully automated	Module A — AI chatbot
Course benefits and learning outcomes	Fully automated	Module A — AI chatbot
Entry without experience or certificates	Fully automated	Module A — AI chatbot
Job opportunities after course completion	Fully automated	Module A — AI chatbot
Follow-up reminders for non-responsive leads	Fully automated	Module A — Scheduler
Group 2 — Enrollment & Payment Administration		
Sending course info templates via WhatsApp/email	Fully automated	Module A + B
Explaining SkillsFuture / PayNow payment modes	Fully automated	Module A + B
SkillsFuture credit balance check (Singpass)	Guided — no gov API	Module A (link + instructions)
Collecting participant details (NRIC, DOB, email)	Fully automated	Module A + B form
Generating accurate invoices (replacing Excel)	Fully automated	Module B — PDF engine
Emailing PDF invoice + SkillsFuture instructions	Fully automated	Module B — Email delivery
Sending PayNow QR code and company UEN	Fully automated	Module B — Email delivery

Requirement	Status	Module
Verifying payment screenshots from participants	AI-assisted + staff confirm	Module C — AI verification
Updating payment records for reconciliation	Fully automated	Module C — Ledger
Coordinating with accounts for receipts & credit notes	Staff portal/Fully automated	Module C — Accountant portal

5. Technical Stack

The system is built as a fully agentic AI workflow, meaning each module operates as an autonomous pipeline: it perceives inputs, makes decisions via an AI reasoning layer, and takes actions without requiring human intervention at each step. The implementation platform of choice is **n8n** — a self-hostable, open-source workflow automation platform that supports native AI agent nodes, LLM tool calling, conditional branching, and webhook-based triggers. The sections below describe each module’s agentic workflow pattern, how it maps to n8n, and the surrounding technology stack.

5.1 Shared Entry Point — WhatsApp Intent Router

All three modules share a single, unified WhatsApp entry point. Every inbound message from a customer — whether an enquiry, an intent to enroll, or a payment screenshot — arrives at the same WhatsApp Business Cloud API webhook. A dedicated **Intent Router** — an n8n AI Agent node powered by Claude — analyses the message content and classifies it into one of three intents: **enquiry**, **enrollment**, or **payment submission**. The router then dispatches the message to the appropriate module webhook automatically, with no manual routing required.

Routing Logic

- **Enquiry intent:** Text messages asking about course details, funding, eligibility, or requiring follow-up — routed to Module A (AI Chatbot & Lead Nurturing).
- **Enrollment intent:** Messages expressing intent to register or sign up, or structured enrollment data — routed to Module B (Enrollment, Invoice & Payment Pipeline).
- **Payment submission intent:** Messages containing an image attachment identified as a payment receipt or screenshot — routed to Module C (Payment Verification & Accounts).

This design means participants never need to navigate to separate links or channels — a single WhatsApp conversation with Q&M is the complete interface for the entire participant journey, from first enquiry through to receipt issuance.

5.2 Agentic Workflow Architecture Overview

An agentic workflow consists of four layers that correspond directly to n8n building blocks: a **Trigger** (perceive), a **Reasoning Agent** (decide), a **Tool / Action Node** (act), and a **Memory / State Store** (remember). The diagram below shows this pattern replicated across all three modules, with a shared Intent Router as the unified entry point for all inbound WhatsApp messages.

Agentic Pattern: WhatsApp Inbound → Intent Router (Claude) → Route to Module A / B / C → AI Agent → Tool Nodes → State Update → Output / Escalate

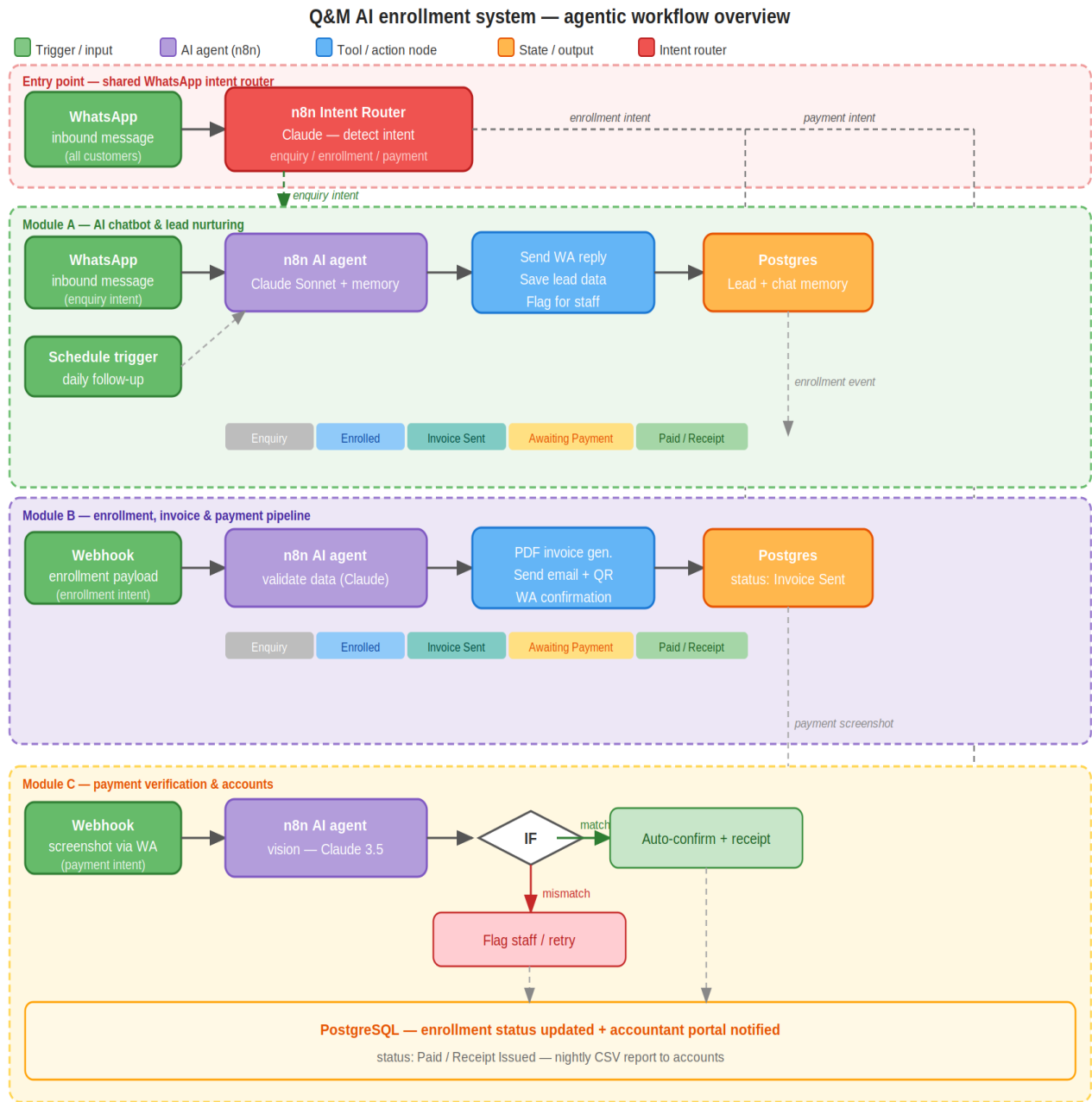


Figure 1: Q&M AI Enrollment System — Agentic workflow overview: shared WhatsApp Intent Router dispatching to Modules A, B, and C

In n8n this pattern maps to: a shared **Intent Router node** (Claude AI — detects enquiry / enrollment / payment intent from every inbound WhatsApp message and dispatches to the appropriate module webhook) feeding into a per-module **Webhook / Schedule Trigger node** feeding into an **AI Agent node** (with an attached Claude or OpenAI Chat Model sub-node), which calls registered **Tool nodes** (HTTP Request, Send Email, PostgreSQL, Google Sheets), and writes

results to a **Memory node** (Simple Memory or Postgres Chat Memory) to maintain conversation and enrollment state across sessions.

5.3 Module A — AI Chatbot & Lead Nurturing (Agentic Conversational Agent)

Agentic Workflow Pattern

Module A is a **Reactive Conversational Agent**. It receives messages pre-classified as enquiry intent by the shared Intent Router. Once routed, Module A is always-on and event-driven: the agent autonomously decides whether to answer from its knowledge base, capture lead data, escalate to staff, or schedule a follow-up. The agent loops until the conversation reaches a terminal state (enrolled, escalated, or timed out).

n8n Implementation Mapping

- **Trigger:** WhatsApp Business Cloud API → n8n Intent Router (Claude) → Module A Webhook. All inbound WhatsApp messages first pass through the shared Intent Router, which detects an enquiry intent and forwards the message payload to the Module A webhook. The node extracts the sender's phone number and message body as workflow variables for the conversational agent.
- **Reasoning Agent:** n8n AI Agent node configured with the Claude Sonnet model (via Anthropic API credential). The system prompt contains the FAQ knowledge base, lead capture instructions, and escalation rules. The agent has access to three registered tools: (1) Save Lead Data, (2) Send WhatsApp Reply, and (3) Flag for Staff Review.
- **Memory / State:** n8n Postgres Chat Memory node keyed on the sender's phone number. This allows the agent to maintain conversation continuity across multiple WhatsApp messages from the same participant.
- **Follow-up Scheduler:** A separate n8n Schedule Trigger workflow runs daily. It queries the leads database (PostgreSQL) for records in "Awaiting Response" status and checks the last-contact timestamp. If the threshold (24h, 72h, or 7d) has elapsed, the workflow invokes the AI Agent to compose and send a contextual follow-up WhatsApp message, then updates the follow-up counter in the database.

Key Technology

- WhatsApp Business Cloud API (Meta) for inbound/outbound messaging
- n8n AI Agent node + Claude Sonnet (Anthropic) as the reasoning LLM
- PostgreSQL (Supabase or self-hosted) for lead records and Postgres Chat Memory

5.4 Module B — Enrollment, Invoice & Payment Pipeline (Agentic Process Automation)

Agentic Workflow Pattern

Module B is a **Sequential Agentic Pipeline**. It is triggered when the shared Intent Router classifies an inbound WhatsApp message as enrollment intent, then autonomously orchestrates a chain of downstream tasks: data validation, invoice generation, document delivery, and status tracking. Unlike Module A it is not conversational — it is a directed, goal-driven pipeline with conditional branching at each step.

n8n Implementation Mapping

- **Trigger:** WhatsApp Business Cloud API → n8n Intent Router (Claude) → Module B Webhook. When a customer's inbound WhatsApp message is classified as enrollment intent, the shared Intent Router forwards the payload to the Module B webhook. The trigger also fires when Module A's chatbot agent calls its "Enroll Participant" tool. The payload includes participant name, NRIC, DOB, email, and preferred course date.

- **AI Data Validation Step:** An n8n AI Agent node (Claude) validates the incoming data — checking NRIC format, date plausibility, and SkillsFuture eligibility based on age — and either proceeds or sends a correction request back to the participant via WhatsApp.
- **Invoice Generation Tool:** An n8n HTTP Request node calls a lightweight PDF microservice (Node.js + Puppeteer or pdfmake, self-hosted on the same server as n8n). The AI agent passes the validated participant data as JSON; the microservice returns a base64-encoded signed PDF invoice. The invoice template includes course fee, SkillsFuture subsidy, net payable, PayNow QR code, and company UEN.
- **Email Delivery Tool:** n8n Send Email node (SMTP or SendGrid) delivers the PDF invoice, SkillsFuture claim instructions PDF, and PayNow details to the participant's email address. The n8n agent waits for the email delivery confirmation before advancing the enrollment record status.
- **State Update:** n8n PostgreSQL node writes the new enrollment record and advances its status to "Invoice Sent". The admin dashboard (a lightweight Next.js or Retool app reading the same database) reflects this in real time.

Key Technology

- n8n workflow with Webhook, AI Agent, HTTP Request, Send Email, and PostgreSQL nodes
- PDF microservice: Node.js + Puppeteer (or pdfmake) deployed alongside n8n
- PostgreSQL (Supabase) as the enrollment state store; Next.js or Retool for admin dashboard

5.5 Module C — Payment Verification & Accounts (Agentic Verification Agent)

Agentic Workflow Pattern

Module C is a **Multimodal Verification Agent**. It receives image inputs (payment screenshots) routed from the shared Intent Router, uses vision-capable LLM reasoning to extract and validate payment details, and then takes a branching action: auto-confirm, flag mismatch for staff, or request resubmission. This is the most autonomous module — a correctly matched payment requires zero human interaction from submission to receipt issuance.

n8n Implementation Mapping

- **Trigger:** WhatsApp Business Cloud API → n8n Intent Router (Claude) → Module C Webhook. When a participant sends a payment screenshot via WhatsApp, the shared Intent Router detects payment submission intent and routes the message to the Module C webhook. The webhook node extracts the sender's phone number and media ID; an n8n HTTP Request node then downloads the image binary using the Meta Graph API and stores it as a base64 string for the vision agent.
- **Multimodal Reasoning Agent:** n8n AI Agent node with a vision-capable model (Claude 3.5 Sonnet or GPT-4o). The agent is given the base64 screenshot and the outstanding invoice amount from the database. It is instructed to extract the transferred amount, recipient UEN, and transaction reference from the screenshot, then compare against the expected values. The agent returns a structured JSON verdict: confirmed, mismatch, or unreadable.
- **Branching Action (n8n IF / Switch node):** If confirmed → update enrollment status to "Paid", trigger receipt generation and email delivery via the PDF microservice, send a WhatsApp confirmation to the participant. If mismatch or unreadable → notify the staff inbox (email or Slack via n8n Send Email / Slack node) with the flagged screenshot and discrepancy details for manual review.
- **Accountant Portal Trigger:** A separate n8n Schedule Trigger workflow runs nightly to generate a summary report of all enrollments by payment status (Paid, Awaiting Payment, Overdue, Credit Note Required) and emails a CSV export to the accounting team. Credit



note approvals from the accountant portal submit a webhook back to n8n, which then auto-generates and emails the credit note PDF.

Key Technology

- n8n workflow with Webhook, HTTP Request (Meta Graph API image download), AI Agent (vision), IF/Switch, PostgreSQL, Send Email, and WhatsApp HTTP nodes
- Claude 3.5 Sonnet (vision) or GPT-4o for multimodal payment screenshot analysis
- Retool or Next.js accountant portal reading PostgreSQL; PDF microservice for receipt and credit note generation

5.6 Full Stack Summary

The table below consolidates all technology components across the three modules.

Layer	Module A	Module B	Module C
Trigger	Intent Router (Claude) → Webhook (enquiry intent routed from WhatsApp inbound)	Intent Router (Claude) → Webhook (enrollment intent routed from WhatsApp inbound, or enrollment event from Module A)	Intent Router (Claude) → Webhook (payment submission intent routed from WhatsApp inbound)
AI Reasoning	n8n AI Agent + Claude Sonnet (conversational + tool calling)	n8n AI Agent + Claude Sonnet (data validation step)	n8n AI Agent + Claude 3.5 Sonnet vision (multimodal screenshot analysis)
Automation Core	n8n (self-hosted) + Schedule Trigger for follow-up sequences	n8n (self-hosted) + PDF microservice (Puppeteer)	n8n (self-hosted) + IF/Switch + Schedule Trigger (nightly reports)
State / Memory	Postgres Chat Memory (per phone number)	PostgreSQL enrollment records (status machine)	PostgreSQL payment records; accountant portal reads same DB
Messaging Channel	WhatsApp Business Cloud API (Meta)	SMTP / SendGrid (email delivery)	WhatsApp (confirmation) + SMTP (receipt); Slack/email (staff alerts)
Infrastructure	n8n self-hosted on a single VPS (e.g. DigitalOcean S\$14/mo droplet or AWS) with Docker Compose. PostgreSQL (Supabase free tier or same VPS). All three module workflows run within the same n8n instance. Estimated running cost: S\$400–600/month total (VPS + Anthropic API + WhatsApp messaging fees).		

6. Implementation Plan

The project is structured across five phases with a total duration of 12 weeks from kick-off to production launch. Timelines are indicative and will be confirmed following the discovery phase.

Phase 1 — Requirements and Discovery (Weeks 1–2)

- Conduct requirements analysis with Q&M's operations team to validate the three-module scope, confirm escalation rules, and finalise the boundary between automated handling and staff intervention
- Inventory and acquire all FAQ content with Q&M staff: SkillsFuture funding eligibility, Mid-Career SkillsFuture Credit rules, course schedule and pricing, learning outcomes, entry requirements, and post-course job pathways
- Review the existing manual workflow end-to-end — WhatsApp message patterns, the current Excel-based invoice template, PayNow QR setup, and accountant reconciliation steps — to define the automation contract for each module
- Confirm the technical stack: n8n hosting (VPS sizing and region), PostgreSQL / Supabase, Anthropic API credentials, WhatsApp Business Cloud API onboarding, PDF microservice approach, and admin dashboard framework (Next.js or Retool)
- Define the PDPA-compliant consent mechanism for participants and the data-retention policy for WhatsApp conversation logs, NRIC, and payment screenshots

Phase 2 — Design (Weeks 3–4)

- Design the WhatsApp conversational flows: chatbot persona and tone, quick-reply menu, lead-capture script, and the 24-hour / 72-hour / 7-day follow-up message templates
- Design the Intent Router classification prompt and the routing rules that dispatch inbound messages between Modules A, B, and C, including ambiguity and fall-back handling
- Design the Module B enrollment pipeline: data-validation schema, invoice PDF template (course fee, SkillsFuture subsidy, net payable, PayNow QR, company UEN), and email templates for invoice and SkillsFuture instructions
- Design the Module C vision-LLM verification prompt for screenshot parsing, the auto-confirm vs. flag thresholds, and the staff review handoff. Payment status report.
- Design the database schema (leads, enrollments, payments, status machine) and the wireframes for the admin dashboard and the accountant portal
- Curate and prepare the seed FAQ knowledge base in a structured format suitable for direct ingestion into the chatbot system prompt

Phase 3 — Development (Weeks 5–8, staggered release)

- Weeks 5–6: Build Module A — the shared WhatsApp Intent Router, the AI Chatbot (Claude Sonnet agent with FAQ knowledge), the lead-capture flow, and the follow-up scheduler; release for early Q&M UAT at the end of Week 6
- Weeks 6–7: Build Module B — the enrollment pipeline, the PDF microservice for invoices (Node.js + Puppeteer), email delivery, and the admin dashboard with real-time status tracking
- Weeks 7–8: Build Module C — the vision-LLM payment verification agent, the accountant portal, and the automated receipt and credit-note PDF generation
- Integrate the three modules behind the shared Intent Router; configure Postgres Chat Memory, the enrollment status machine, and the staff escalation channels (email and/or Slack)
- Deploy n8n, PostgreSQL, and the PDF microservice on the production VPS with configure automated database backups, log retention, and basic uptime monitoring

Phase 4 — Testing and UAT (Weeks 9–10)

- End-to-end functional testing: validate the full participant journey from first enquiry through enrollment, payment submission, and receipt issuance against a defined test pack of representative scenarios
- Intent Router accuracy evaluation: measure classification precision and recall across enquiry, enrollment, and payment-submission messages, including ambiguous and adversarial edge cases; tune the router prompt accordingly
- Vision verification testing: benchmark Module C against a labelled set of real and synthetic payment screenshots (matched, mismatched, blurred, multi-payment, unreadable) and tune the auto-confirm confidence thresholds
- User Acceptance Testing: Q&M operations and accounts staff run live conversations against the staging environment using realistic scenarios, raising defects and clarifications which are triaged and resolved in-sprint
- Security and PDPA compliance audit: validate that NRIC and payment data are encrypted at rest, no personally identifiable information is retained beyond the policy window, API access controls and secrets management are correctly configured, and the consent text is in place

Phase 5 — Deployment and Post-Launch (Weeks 11–12)

- Cut over from the staging WhatsApp number to Q&M's production WhatsApp Business number; configure DNS, SSL certificate, and the public webhook endpoint; verify Meta webhook delivery in production
- Conduct Q&M staff onboarding: chatbot capabilities, escalation handling, the admin dashboard, the accountant portal, and the procedure for updating FAQ content after handover
- Issue handover documentation: system architecture, operations run-book, content-update guide, backup and restore procedure, and the PDPA compliance dossier
- Activate post-launch monitoring; commence 30-day hyper-care support; report on KPIs at the 4-week and 8-week post-launch milestones

Note: The schedule above is indicative. Final timelines will be confirmed following the discovery phase and are subject to change in accordance with Q&M's operational priorities and the WhatsApp Business API onboarding timeline.

**Meta whatsapp API will be use for the whatsapp server. However, it take time to register for the business whatsapp hence we will do a customer whatsapp for the purpose of development or proof of concept. It can be migrate to the meta whatsapp with minimum effect.*

7. Delivery Items

The project delivers a working, production-deployed automation system together with the supporting content, documentation, and operational artefacts required for Q&M to operate it independently after handover.

7.1 Software Deliverables

- Self-hosted n8n instance with all four workflow components deployed and operational: the shared WhatsApp Intent Router, Module A (Chatbot & Lead Nurturing), Module B (Enrollment, Invoice & Payment Pipeline), and Module C (Payment Verification & Accounts)
- PostgreSQL database (Supabase or self-hosted) with leads, enrollments, payments, and chat-memory schemas, including the status state machine for the enrollment lifecycle
- PDF microservice (Node.js + Puppeteer) for invoice, receipt, and credit-note generation, deployed alongside n8n. The option may change to the equivalent upon the development situation.

- Admin dashboard (Next.js or Retool) for Q&M operations staff to view enrollments, statuses, and escalations
- Accountant portal with password-protected access for payment status review, pending receipts, overdue accounts, and credit-note approval
- WhatsApp Business Cloud API integration configured against Q&M's production WhatsApp Business number

7.2 Content Deliverables

- Curated FAQ knowledge base for the chatbot, covering SkillsFuture funding eligibility, Mid-Career Credit rules, course benefits, entry requirements, and post-course job pathways
- Invoice, receipt, and credit-note PDF templates branded for Q&M and approved during the design phase
- Email templates for SkillsFuture claim instructions, PayNow payment instructions, enrollment confirmation, and receipt delivery
- WhatsApp follow-up message templates for the 24-hour, 72-hour, and 7-day nurture sequence

7.3 Documentation Deliverables

- System architecture document covering the Intent Router, the three modules, data flow, and integration points
- Operations run-book: start/stop procedures, common-failure recovery, log review, and the database backup and restore procedure
- Staff user guide for the admin dashboard, the accountant portal, and the staff escalation handling workflow
- Content-update guide explaining how Q&M staff can edit the FAQ knowledge base and the email/PDF templates after handover

7.4 Operational Deliverables(for proof of concept or MVP)

- Production deployment on a Q&M-owned VPS, automated database backups, log retention, and uptime monitoring
- Training sessions: one for the operations team (chatbot, dashboard, escalations) and one for the accounting team (accountant portal, receipt and credit-note workflow)
- 30-day support on the agreed deliverables and application functions

8. Success Metrics (Key Performance Indicators)

The KPIs below will be tracked from go-live to measure the system's automation effectiveness, quality, participant experience, and operational stability. Metrics will be reported at the 4-week and 8-week post-launch milestones and subsequently on a monthly basis. Baselines for comparison metrics will be established during the discovery phase from Q&M's current manual workflow.

KPI	Definition	Target
System Quality & Accuracy		
Intent Router Classification Accuracy	Proportion of inbound messages correctly routed to Module A, B, or C versus a sampled human-labelled ground truth	≥95%
Chatbot FAQ Answer Accuracy	Proportion of chatbot answers validated as factually accurate from Q&M's FAQ content	≥95%

KPI	Definition	Target
Invoice & Receipt Data Accuracy	Proportion of generated invoices and receipts with zero data errors in name, NRIC, course date, and fee fields	100%
Payment Verification False-Match Rate	Proportion of auto-confirmed payments later found to be incorrect on subsequent accounting reconciliation	0%

9. Assumptions & Exclusions

Assumptions

- Q&M holds or will obtain a WhatsApp Business API account (Meta Cloud API — available directly through Meta, no third-party BSP required)
- Q&M staff will provide all FAQ content, course details, pricing, and SkillsFuture funding information for chatbot training during the discovery phase (Weeks 1–2)
- The project developer will have access to a staging WhatsApp number for development and testing
- The developed application will not be integrating with current Q&M's existing accounting software or other currently operating systems.

Exclusions

- Direct integration with SingPass or SkillsFuture government APIs — excluded on national data security grounds; replaced with guided self-service instructions
- Payment gateway integration (Stripe, PayPal, etc.) — the system supports PayNow and SkillsFuture claims only, consistent with Q&M's current payment methods
- The system is fully web-based; participant interaction occurs through WhatsApp and a email
- Ongoing content updates to the FAQ knowledge base after handover — these are managed by Q&M staff via an admin content editor

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10. WhatsApp Command Schema — Stateless Design

WhatsApp webhook calls are stateless: each inbound message is an independent event with no memory of prior turns. The backend cannot ask follow-up questions. Every command must therefore carry all required data in a single message. If data is missing or malformed, the bot returns an error response with the exact format the user must resend. No session, no conversation history, no multi-turn dialogue.

Fields within a command are separated by a pipe character (|). The user types the command keyword, then all required fields in order. Commands that need no data (e.g. /courses, /mystatus) work as single keywords.

10.1 Command Registry

Command	Full syntax (single message)	Description	Module
/help	/help	Show all commands and their syntax	System
/courses	/courses	List all available courses	Module A
/fees	/fees [course]	Fee & SkillsFuture info for one course	Module A
/schedule	/schedule [course]	Upcoming intake dates for one course	Module A
/sfc	/sfc	General SFC info + portal link (no personal check)	Module A
/enroll	/enroll [course] [name] [NRIC] [email]	Enroll: all 4 fields in one message	Module B
/mystatus	/mystatus	Lookup enrollment status by phone number	Module B
/invoice	/invoice	Resend invoice to email on record	Module B
/pay	/pay + [screenshot attached]	Submit PayNow screenshot as image caption	Module C
/receipt	/receipt	Resend receipt to email on record	Module C

10.2 Module A — Enquiry Commands

Module A commands require at most one parameter (the course name or number). If the parameter is omitted, the bot returns an error with the expected format.

/fees [course] and /schedule [course]

Inbound: /fees Infection Control

Outbound: Full fee breakdown, SkillsFuture subsidy, and /enroll format hint

Error (no course given): Bot returns expected format: /fees [course name or number]

/sfc — general SkillsFuture info only, no personal data collected

Per the design exclusion in Section 3, direct Singpass or SkillsFuture API integration is excluded. /sfc provides general eligibility criteria and a link to myskillsfuture.gov.sg only.

Inbound: /sfc

Outbound: Eligibility criteria, credit amounts, portal URL, and /enroll format hint

10.3 Module B — Enrollment Commands

The /enroll command requires all four enrollment fields in a single message. Fields are pipe-separated. The backend validates each field; if any is missing or invalid, it returns a specific error message identifying the problem and the exact retry format.

/enroll [course] | [full name] | [NRIC] | [email]

Inbound (complete): /enroll Infection Control | Tan Wei Ling | S8512345A | tanwl@email.com

Outbound (success): Enrollment confirmation with course, date, name, fee, invoice number sent to email

Inbound (missing email): /enroll Infection Control | Tan Wei Ling | S8512345A

Outbound (error): Identifies missing field: email address. Provides full retry format and example.

Inbound (invalid NRIC): /enroll Infection Control | Tan Wei Ling | 851234 | tanwl@email.com

Outbound (error): Identifies NRIC format error. States expected pattern: S/T/F/G + 7 digits + letter.

/mystatus and /invoice

Inbound: /mystatus

Outbound: Enrollment status, course, date, invoice number, and next action hints — identified by sender phone number, no parameters needed

10.4 Module C — Payment Commands



WhatsApp supports image messages with a text caption. The user attaches their PayNow screenshot and types /pay as the caption — both arrive in the same webhook payload. The backend reads the image from the media field and the command from the caption. Sending /pay as plain text without an image attachment returns an error.

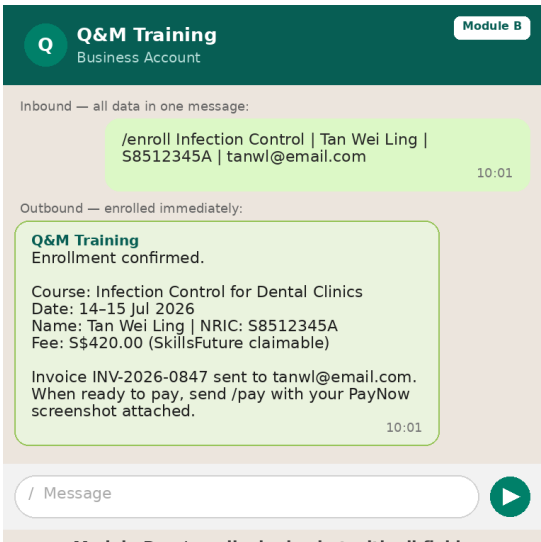
/pay — screenshot attached, /pay as caption

- Inbound: Image message with PayNow screenshot; caption text = /pay
- Outbound (verified): Confirmed amount, date, reference number; receipt sent to email
- Outbound (mismatch): Detected amount does not match invoice. Bot states both amounts and asks to resend correct screenshot with /pay caption.
- Outbound (text-only /pay): No screenshot received. Explains how to attach screenshot with /pay as caption.

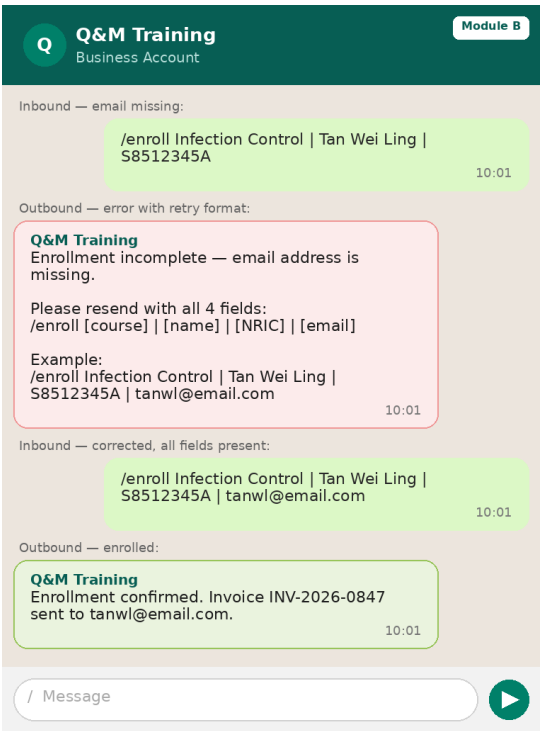
10.5 WhatsApp Chat Mockups — Stateless Design

The following mockups show the single-shot command pattern. Each scenario is a complete atomic interaction: one inbound message contains all required data, and one outbound response either confirms success or states precisely what needs to be corrected and resent.

Module B — /enroll: success (all fields) and error/retry

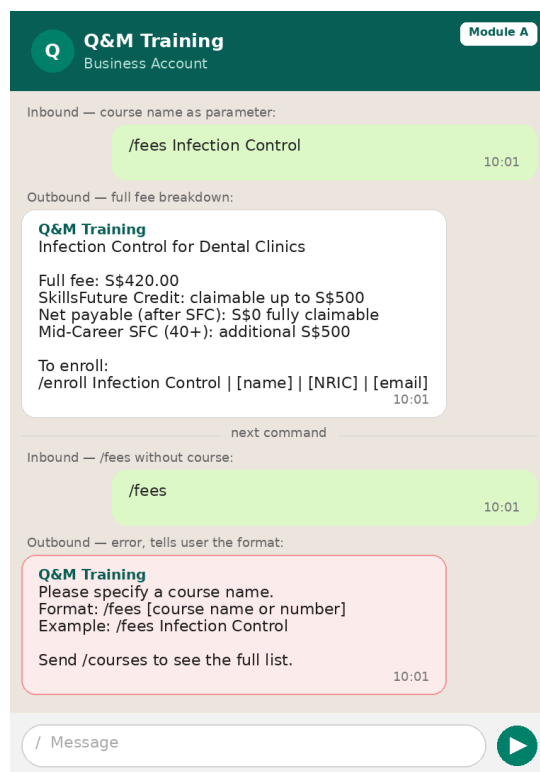


Module B — /enroll: single-shot with all fields



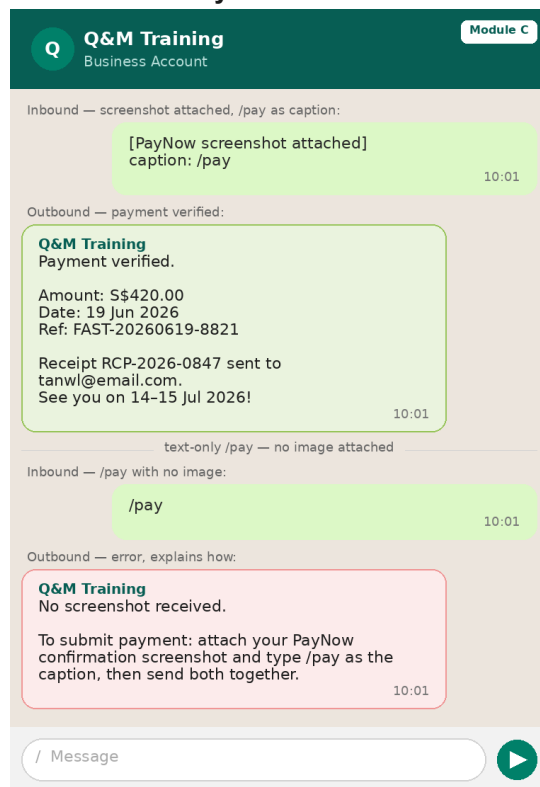
Module B — /enroll: missing field error and retry

Module A — /fees with course parameter and error when omitted



Module A — /fees and /schedule: single-shot with course parameter

Module C — /pay with screenshot and text-only error



Module C — /pay: screenshot + /pay caption in one message

Module B — /mystatus: phone-number lookup, no parameters

Q

Q&M Training

Business Account

Module B

Inbound — command only, no params needed:

/mystatus

10:01

Outbound — status from phone number lookup:

Q&M Training

Your enrollment status:

Course: Infection Control for Dental Clinics

Date: 14–15 Jul 2026

Status: Awaiting payment

Invoice: INV-2026-0847 — S\$420.00

To pay: send /pay with PayNow screenshot attached.

To resend invoice: send /invoice

10:01

no record found

Outbound — not enrolled:

Q&M Training

No enrollment found for your number.

To enroll, send:

/enroll [course] | [name] | [NRIC] | [email]

10:01

/ Message

Module B — /mystatus: phone-number lookup, no parameters